

Quiz details

Quiz instructions:

This is the practice quiz for the Week 5 module content, which includes probability models and their interpretation, and the central limit theorem. There is no time limit on this quiz, and you can take it as many times as you wish.

To answer the questions in this quiz, you will need to use the **power up**  (https://raw.githubusercontent.com/bradduthie/SCIU4T4/main/data/power_up.csv) dataset from the Week 4 lab (right click the link and select 'Save link as...', then save the file as 'power_up.csv'). This dataset is explained in **Chapter 16**  (https://bradduthie.github.io/SCIU4T4/Chapter_16.html) of the lab workbook. Briefly, the dataset includes observations from players of a video game about environmental justice and hydropower. Each player has an ID (Player_ID), operating system (OS), dam size choice (Dam_size), Biodiversity investment (Biodiversity), Community investment (Community), Energy investment (Energy), and final score (Score).

Note that in an actual test or exam, you will not lose the full points for a question if you report an incorrect number of decimals or significant figures (you will only lose half the total points for the question).

☒ Show question details

Question 1 1 pts

What is the standard error of player score across all players? Report your answer to 2 significant figures, and do not include units in your answer.

2.6

Correct answer comments

To find the standard error, navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Score' to the Variables box. Scroll down to the 'Statistics' drop-down menu and make sure that 'Std error of Mean' is checked. The Descriptives table that appears to the right will include a row for the standard error. The value will be 2.59, which is 2.6 when rounded to 2 significant figures.

Wrong answer comments

To find the standard error, navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Score' to the Variables box. Scroll down to the 'Statistics' drop-down menu and make sure that 'Std error of Mean' is checked. The Descriptives table that appears to the right will include a row for the standard error. The value will be 2.59, which is 2.6 when rounded to 2 significant figures.

Question 2 1 pts

What is the standard error of player scores for players who choose a large dam? Report your answer to 2 decimal places, and do not include units in your answer.

3.81

Correct answer comments

To find the standard error, navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Score' to the Variables box, then the variable 'Dam_size' to the 'Split by' box. Scroll down to the 'Statistics' drop-down menu and make sure that 'Std error of Mean' is checked. The Descriptives table that appears to the right will include a row for the standard error. The value will be 3.808, which is 3.81 when rounded to 2 decimal places.

Wrong answer comments

To find the standard error, navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Score' to the Variables box, then the variable 'Dam_size' to the 'Split by' box. Scroll down to the 'Statistics' drop-down menu and make sure that 'Std error of Mean' is checked. The Descriptives table that appears to the right will include a row for the standard error. The value will be 3.808, which is 3.81 when rounded to 2 decimal places.

Question 3 1 pts

What are the 95% confidence intervals around mean player score? Report your answer to 1 decimal place.

Lower 95% CI: [lowerCI]

Upper 95% CI: [upperCI]

Show answers for

upperCI



101.1

Correct answer comments

To find the 95% confidence intervals, navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Score' to the Variables box. Scroll down to the 'Statistics' drop-down menu and make sure that 'Confidence interval for the Mean' is checked. The Descriptives table that appears to the right will include a rows for '95% CI mean lower bound' and '95% CI mean upper bound'. The values will be 90.76 for the lower CI and 101.1 for the upper CI. When rounded to 1 decimal place, the answer is therefore Lower 95% CI = 90.8 and Upper 95% CI = 101.1.

Wrong answer comments

To find the 95% confidence intervals, navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Score' to the Variables box. Scroll down to the 'Statistics' drop-down menu and make sure that 'Confidence interval for the Mean' is checked. The Descriptives table that appears to the right will include a rows for '95% CI mean lower bound' and '95% CI mean upper bound'. The values will be 90.76 for the lower CI and 101.1 for the upper CI. When rounded to 1 decimal place, the answer is therefore Lower 95% CI = 90.8 and Upper 95% CI = 101.1.

Question 4 1 pts

In Question 3, what distribution did Jamovi use to calculate the 95% confidence intervals?

- ☐ Poisson distribution
- ☐ t-distribution

- ☐ normal distribution
- ☐ binomial distribution
- ☐ uniform distribution
- ☐ None of the above

Correct answer comments

Jamovi uses a t-distribution to calculate confidence intervals in the Descriptives menu. This is mentioned in [Chapter 18](https://bradduthie.github.io/SCIU4T4/Chapter_18.html) [⇒ \(https://bradduthie.github.io/SCIU4T4/Chapter_18.html\)](https://bradduthie.github.io/SCIU4T4/Chapter_18.html) of the workbook and in the [first exercise](https://bradduthie.github.io/SCIU4T4/Chapter_19.html#confidence-intervals-with-distraction) [⇒ \(https://bradduthie.github.io/SCIU4T4/Chapter_19.html#confidence-intervals-with-distraction\)](https://bradduthie.github.io/SCIU4T4/Chapter_19.html#confidence-intervals-with-distraction) of the practical for Week 5. Another sensible answer would be 'normal distribution', which could be used to calculate CIs (especially for large sample sizes), but statistical software such as Jamovi and R will always use the t-distribution to correct for finite sample size.

Wrong answer comments

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Question 5 1 pts

What are the 95% confidence intervals around the proportion of players who choose a medium dam size? Report your answer to 2 decimal places.

Lower 95% CI: [lowerCI]

Upper 95% CI: [upperCI]

Show answers for

upperCI



0.26

Correct answer comments

To do this, first navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Dam_size' to the Variables box, then check the 'Frequency tables' box below. A table of frequencies will appear on the panel to the right with counts for large (40), medium (13), and small (21) dam sizes. To get the proportion of people who chose a medium dam size, divide 13 by the total 40 + 13 + 21 dams, $13 / 74 = 0.1756757$. Next, use the formula below introduced in Chapter 17.2, To do this, first navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Dam_size' to the Variables box, then check the 'Frequency tables' box below. A table of frequencies will appear on the panel to the right with counts for large (40), medium (13), and small (21) dam sizes. To get the proportion of people who chose a medium dam size, divide 13 by the total 40 + 13 + 21 dams, $13 / 74 = 0.1756757$. Next, use the formula below introduced in Chapter 17.2 to find the standard error of the proportion (p),

$$SE(p) = \sqrt{\frac{p(1-p)}{N}}.$$

The calculation is therefore,

$$SE(p) = \sqrt{\frac{0.1756757(1-0.1756757)}{74}} = 0.04423734.$$

With this calculation made, we can use the general formula for lower and upper confidence intervals.

$$LCI = \bar{x} - (z \times SE),$$

$$UCI = \bar{x} + (z \times SE).$$

The z-score associated with 95% confidence intervals is 1.96. This was also mentioned in [Chapter 17.1](#) [⇒ \(https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis\)](https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis) of the workbook; you can use the [interactive application](#) [⇒ \(https://bradduthie.shinyapps.io/zandp/\)](https://bradduthie.shinyapps.io/zandp/) to visualise it more clearly. If we put our values into the above equations, we get the following,

$$LCI = 0.1756757 - (1.96 \times 0.04423734) = 0.08897051,$$

$$UCI = 0.1756757 + (1.96 \times 0.04423734) = 0.2623809.$$

When rounding these equations to 2 decimal places, we get the final answer of LCI = 0.09 and UCI = 0.26.

Wrong answer comments

To do this, first navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Dam_size' to the Variables box, then check the 'Frequency tables' box below. A table of frequencies will appear on the panel to the right with counts for large (40), medium (13), and small (21) dam sizes. To get the proportion of people who chose a medium dam size, divide 13 by the total 40 + 13 + 21 dams, 13 / 74 = 0.1756757. Next, use the formula below introduced in Chapter 17.2, To do this, first navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Dam_size' to the Variables box, then check the 'Frequency tables' box below. A table of frequencies will appear on the panel to the right with counts for large (40), medium (13), and small (21) dam sizes. To get the proportion of people who chose a medium dam size, divide 13 by the total 40 + 13 + 21 dams, 13 / 74 = 0.1756757. Next, use the formula below introduced in Chapter 17.2 to find the standard error of the proportion (p),

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$$UCI = 0.1756757 + (1.96 \times 0.04423734) = 0.2623809.$$

When rounding these equations to 2 decimal places, we get the final answer of LCI = 0.09 and UCI = 0.26.

Question 6 1 pts

What are the 99% confidence intervals around mean player score for players who use an iPhone? Report your answer to 1 decimal place.

Lower 99% CI: [lowerCI]

Upper 99% CI: [upperCI]

Show answers for

upperCI



103.1

Correct answer comments

To find the 95% confidence intervals, navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Score' to the Variables box and put 'OS' in the 'Split by' box. Scroll down to the 'Statistics' drop-down menu and make sure that 'Confidence interval for the Mean' is checked and '99' is in the box next to it. The Descriptives table that appears to the right will include a rows for '99% CI mean lower bound' and '99% CI mean upper bound', split by OS. The values will be 67.32 for the lower CI and 103.1 for the upper CI. When rounded to 1 decimal place, the answer is therefore Lower 99% CI = 67.3 and Upper 99% CI = 103.1.

Wrong answer comments

To find the 95% confidence intervals, navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Score' to the Variables box and put 'OS' in the 'Split by' box. Scroll down to the 'Statistics' drop-down menu and make sure that 'Confidence interval for the Mean' is checked and '99' is in the box next to it. The Descriptives table that appears to the right will include a rows for '99% CI mean lower bound' and '99% CI mean upper bound', split by OS. The values will be 67.32 for the lower CI and 103.1 for the upper CI. When rounded to 1 decimal place, the answer is therefore Lower 99% CI = 67.3 and Upper 99% CI = 103.1.

Question 7 1 pts

What would the 95% confidence intervals around the mean player score for players

who use iPhone be if they were calculated with a normal distribution, rather than a t-distribution? Report your answer to 1 decimal place.

Lower 95% CI: [lowerCI]

Upper 95% CI: [upperCI]

Show answers for

97.3

Correct answer comments

To answer this question, you first needed to find the mean and standard error of scores for players who use iPhone.

Mean = 85.23

Standard error = 6.179

You can do this in the Descriptives menu of Jamovi, splitting Scores by OS. Next, you need to use the formula for calculating confidence intervals, first introduced in [Chapter 17.1](https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis) (https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis),

$$LCI = \bar{x} - (z \times SE),$$

$$UCI = \bar{x} + (z \times SE).$$

Note that the z-scores associated with 95% of the probability density in a standard normal distribution (i.e., distribution with mean of 0 and standard deviation of 1) is 1.96, so these are the ones that we need to use in the formula above (i.e., we need to multiply the SE by 1.96),

$$LCI = 85.23 - (1.96 \times 6.179) = 73.11916,$$

$$UCI = 85.23 + (1.96 \times 6.179) = 97.34084.$$

When rounded to 1 decimal place, the correct answer is therefore LCI = 73.1 and UCI = 97.3. Note that to get the correct z-score to do this (in addition to getting it from [Chapter 17.1](https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis) (https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis), you could have used the distrACTION module, or the [interactive application](https://bradduthie.shinyapps.io/zandp/) (<https://bradduthie.shinyapps.io/zandp/>) from [Chapter 17.1](https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis) (https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis).

Wrong answer comments

To answer this question, you first needed to find the mean and standard error of scores for players who use iPhone.

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You can do this in the Descriptives menu of Jamovi, splitting Scores by OS. Next, you need to use the formula for calculating confidence intervals, first introduced in [Chapter 17.1](https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis) (https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis),

$$LCI = \bar{x} - (z \times SE),$$

$$UCI = \bar{x} + (z \times SE).$$

Note that the z-scores associated with 95% of the probability density in a standard normal distribution (i.e., distribution with mean of 0 and standard deviation of 1) is 1.96, so these are the ones that we need to use in the formula above (i.e., we need to multiply the SE by 1.96),

$$LCI = 85.23 - (1.96 \times 6.179) = 73.11916,$$

$$UCI = 85.23 + (1.96 \times 6.179) = 97.34084.$$

When rounded to 1 decimal place, the correct answer is therefore LCI = 73.1 and UCI = 97.3. Note that to get the correct z-score to do this (in addition to getting it from [Chapter 17.1](https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis) (https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis), you could have used the distrACTION module, or the [interactive application](https://bradduthie.shinyapps.io/zandp/) (<https://bradduthie.shinyapps.io/zandp/>) from [Chapter 17.1](https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis) (https://bradduthie.github.io/SCIU4T4/Chapter_17.html#normal-distribution-cis).

Question 8 1 pts

What would the 95% confidence intervals around the mean player score for players who use iPhone be if they were calculated with a t-distribution, rather than a normal distribution? Report your answer to 1 decimal place.

Lower 95% CI: [lowerCI]

Upper 95% CI: [upperCI]

Show answers for

upperCI



98.3

Correct answer comments

To find the 95% confidence intervals, navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Score' to the Variables box, and put 'OS' in the 'Split by' box. Scroll down to the 'Statistics' drop-down menu and make sure that 'Confidence interval for the Mean' is checked. The Descriptives table that appears to the right will include a rows for '95% CI mean lower bound' and '95% CI mean upper bound'. The values will be 72.19 for the lower CI and 98.27 for the upper CI. When rounded to 1 decimal place, the answer is therefore Lower 95% CI = 72.2 and Upper 95% CI = 98.3.

Wrong answer comments

To find the 95% confidence intervals, navigate to 'Exploration', then 'Descriptives' in the Jamovi toolbar. Next, move the variable 'Score' to the Variables box, and put 'OS' in the 'Split by' box. Scroll down to the 'Statistics' drop-down menu and make sure that 'Confidence interval for the Mean' is

checked. The Descriptives table that appears to the right will include a rows for '95% CI mean lower bound' and '95% CI mean upper bound'. The values will be 72.19 for the lower CI and 98.27 for the upper CI. When rounded to 1 decimal place, the answer is therefore Lower 95% CI = 72.2 and Upper 95% CI = 98.3.

Question 9 1 pts

Given the answers to Question 7 and Question 8, which distribution (t- or normal) appears to express more confidence intervals around the mean?

- ☐ normal distribution because the confidence intervals are narrower
- ☐ normal distribution because the confidence intervals are wider
- ☐ t distribution because the confidence intervals are narrower
- ☐ t distribution because the confidence intervals are wider
- ☐ It is impossible to determine this

Correct answer comments

The correct answer here is A. The normal distribution expresses more confidence because the confidence intervals (i.e., the difference between the LCI and UCI) is smaller when using the z-scores rather than t-scores (as calculate in Jamovi).

Wrong answer comments

The correct answer here is A. The normal distribution expresses more confidence because the confidence intervals (i.e., the difference between the LCI and UCI) is smaller when using the z-scores rather than t-scores (as calculate in Jamovi).

Question 10 1 pts

Which of the following answers is the best explanation for why the confidence intervals from Question 7 versus Question 8 are different?

- ☐ Confidence intervals calculated from the t-distribution, but not the normal distribution, assume a very large sample size
- ☐ Confidence intervals calculated from the normal distribution, but not the t-distribution, assume a very large sample size
- ☐ Confidence intervals calculated from the t-distribution, but not the normal distribution, assume independence among player scores
- ☐ Confidence intervals calculated from the normal distribution, but not the t-distribution, assume independence among player scores
- ☐ Confidence intervals calculated from the t-distribution, but not the normal distribution, pools the standard error across the entire dataset.
- ☐ Confidence intervals calculated from the normal distribution, but not the t-distribution, pools the standard error across the entire dataset

Correct answer comments

The normal distribution assumes a very large (technically infinite) sample size, and this causes our confidence in the mean to be over-estimated. To correct for a small sample size, we need to use the t-distribution, which is what Jamovi does automatically. See [Chapter 18](#) ➞ https://bradduthie.github.io/SCIU4T4/Chapter_18.html for more details.

Wrong answer comments

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